

Parameter	Resistive	Inductive	Capacitive	R-L Series	R-C Series	R-L-C Series	
						$X_L > X_C$	$X_C > X_L$
Equivalent Circuit							
Volatge	$V_m \sin \omega t$	$V_m \sin \omega t$	$V_m \sin \omega t$	$V_m \sin \omega t$	$V_m \sin \omega t$	$V_m \sin \omega t$	$V_m \sin \omega t$
Current	$I_m \sin \omega t$	$I_m \sin(\omega t - \frac{\pi}{2})$	$I_m \sin(\omega t + \frac{\pi}{2})$	$I_m \sin(\omega t - \phi)$	$I_m \sin(\omega t + \phi)$	$I_m \sin(\omega t - \phi)$	$I_m \sin(\omega t + \phi)$
Power	$P = VI$	0 (Zero)	0 (Zero)	$P = VICOS\phi$	$P = VICOS\phi$	$P = VICOS\phi$	$P = VICOS\phi$
Power Factor	Unity (1)	0 (Lagging)	0 (Leading)	$COS\phi = \frac{R}{Z}$	$COS\phi = \frac{R}{Z}$	$COS\phi = \frac{R}{Z}$	$COS\phi = \frac{R}{Z}$
Impedance	$R = \text{Resistance}$	$X_L = \text{Inductive Reactance}$	$X_C = \text{Capacitive Reactance}$	$Z = \sqrt{R^2 + X_L^2}$	$Z = \sqrt{R^2 + X_C^2}$	$Z = \sqrt{R^2 + (X_L - X_C)^2}$	$Z = \sqrt{R^2 + (X_C - X_L)^2}$
Phasor Daigram							
Voltage Triangle	x	x	x				
Impedance Triangle	x	x	x				
Waveform							